SEPARABILITY ANALYSIS OF STRESS-RELATED PHYSIOLOGICAL PATTERNS

A Presentation by
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AGENDA OVERVIEW

- Background and Novelty
- Research Question
- Dataset and Process
- Algorithms
- Dataset Analysis
- Q&A

BACKGROUND

WHAT IS STRESS?

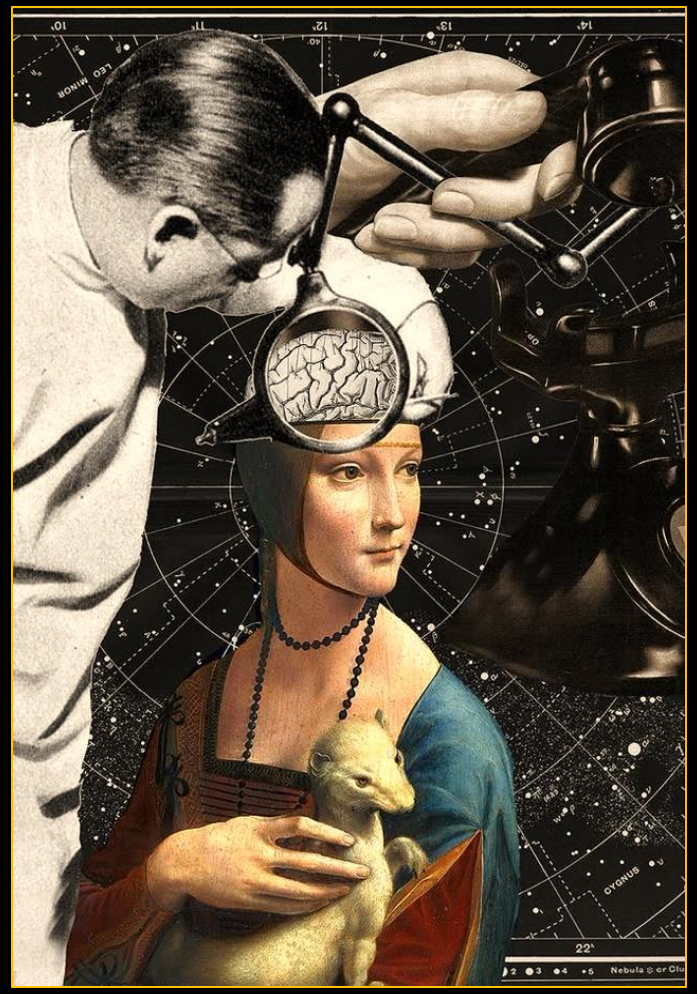
Stress is a **natural, involuntary response** to perceived demands and/or threats.

Triggers physiological, emotional, and cognitive responses.

Short-term stress (acute) can be adaptive, but prolonged stress (chronic) can be harmful over time.

Chronic stress can lead to serious health issues, and loss of quality of life.

BACKGROUND & NOVELTY



- Analyzing stress-related physiological patterns has potential implications for early awareness and preventative health monitoring

- Wearable devices **enable continuous collection of physiological signals** such as heart rate and electrodermal activity, so we can analyze patterns associated with stress-related responses.

- Other studies focus on the predictability accuracy of stress, while this project looks at the interpretations of the recognized patterns that may indicate elevated stress-related risk

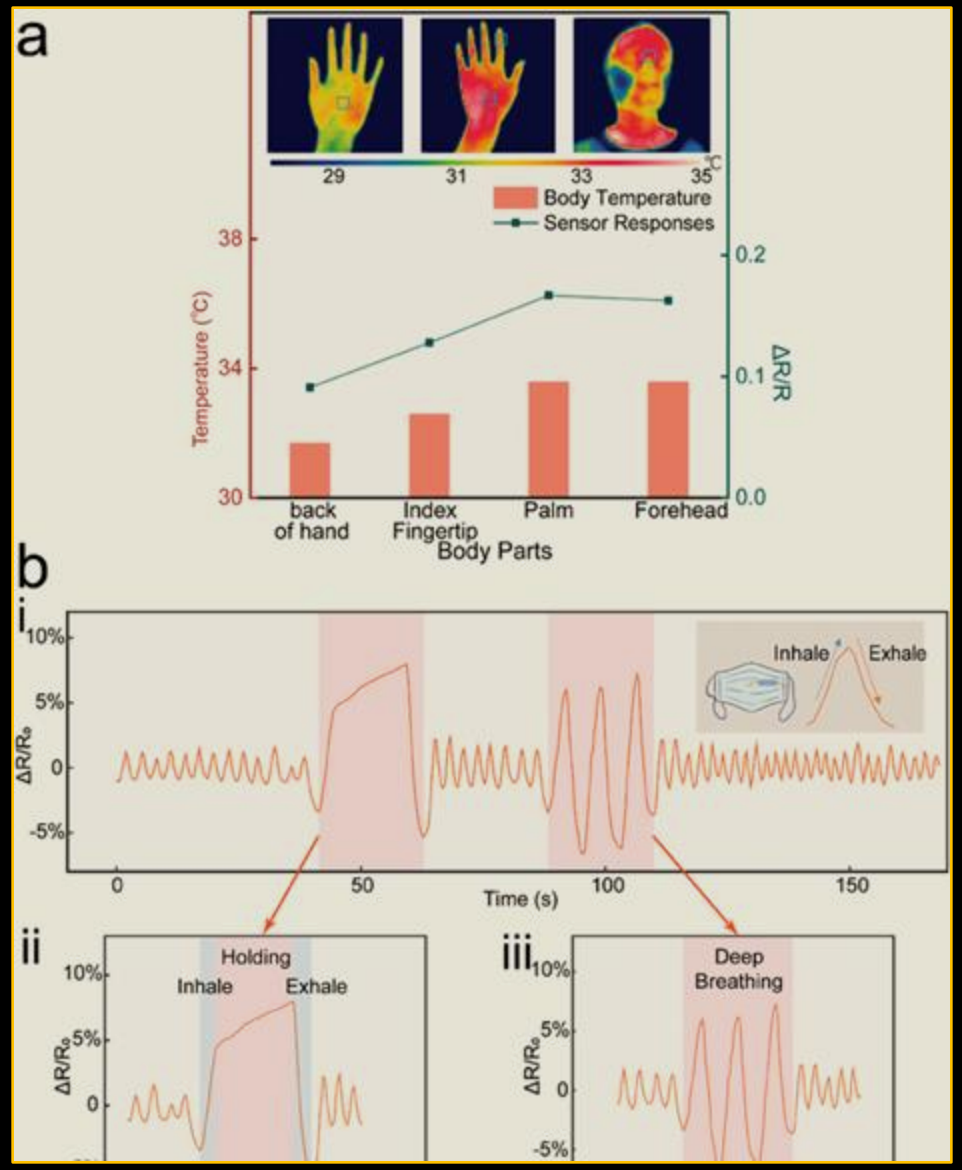
HOW DO DIFFERENT PHYSIOLOGICAL FEATURE REPRESENTATIONS AND PATTERN RECOGNITION METHODS AFFECT THE **SEPARABILITY AND DECISION BEHAVIOR** OF STRESS-RELATED RISK STATES IN **WEARABLE DATA?**

CLARIFICATION

Big picture: Studying the found patterns in stress physiology from the dataset

Risk State: Stress-associated patterns
(This is not about diagnosing)

Ex) an elevated risk state means that the pattern is near a stress region



DATASET

Wearable Stress and Affect Detection (WESAD)

Multimodal wearable physiology dataset

Many participants, multiple recording sessions

Signals: ECG (HR/HRV), EDA, skin temperature, motion

Conditions: baseline (low stress) and stress-inducing tasks

Used to analyze whether stress-associated physiological patterns are separable from baseline patterns

TEMP_mean	TEMP_std	label	subject_id
33.70098456	0.0226305623277865	2	S4
33.69958846	0.027417795924463178	2	S4
33.699434620000005	0.02069262108538393	2	S4
33.69988968	0.016013598814681326	2	S4
33.6883424	0.02292103541042073	2	S4
33.69964954	0.020692471581013144	2	S4
33.70407282	0.015381199785598564	2	S4
33.69673832	0.0175443275959347	2	S4
33.69782901999999	0.025234082000900857	2	S4
33.68573604	0.019262177012572778	2	S4
33.69349678	0.019063950854298493	2	S4
33.699466580000006	0.025917808700590832	2	S4
33.71119806	0.02026942889403178	2	S4
33.71617134	0.026701287243761897	2	S4
34.317123360000004	0.02683195107403142	1	S5
34.32981	0.02136157952579186	1	S5
34.327917400000004	0.026819458621208023	1	S5
34.38906494	0.02827112927973378	1	S5
34.42668446	0.02909404492346426	1	S5
34.43063316	0.02525716381330776	1	S5

DATASET PROCESSING

- Collected raw multimodal signals
- Filtered to binary labels
  - (Stress vs Baseline)
- Segmented data into windows of 50
  - Each window represents a time interval
  - ECG, EDA, TEMP from each window

Raw Signals → Windowing → Feature Extraction → Dataset

ALGORITHMS USED

- **K-Nearest Neighbors (K-NN)**: whether "similar physiological patterns" cluster by condition
- **Support Vector Machines (linear and RBF):** Linear separability, and RBF for overlap boundaries
- **Naïve Bayes:** test whether stress vs baseline features follow separable distributions
- **Class-weighted SVM:** explore risk-oriented decision behavior
- **All models and experiments were implemented in Python**

```
import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, classification_report
```

ANALYSIS

RESULT SUMMARY

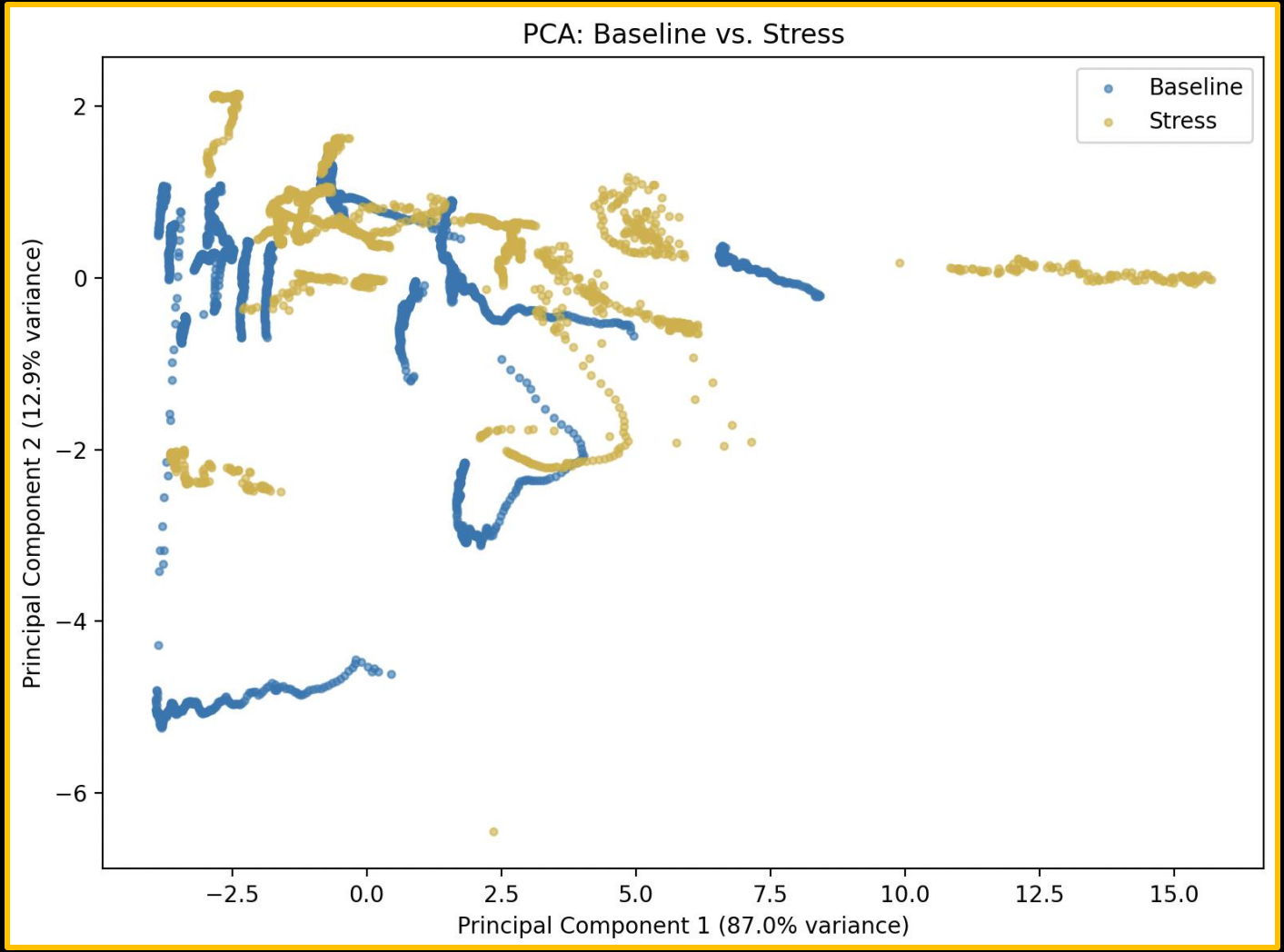
- Classification accuracy ~65% across models
- Models heavily favored baseline predictions
- Stress detection recall remained low
- Using EDA features, we observed limited detection of stress
- Recall remains low, indicating weak separability.
- Stress detection recall: 8%
- The model is accurate overall, but fails at the task we actually care about – detecting stress

```
--- EDA Only ---
Accuracy: 0.6780626780626781
[[228   1]
 [112  10]]
              precision    recall

           1       0.67      1.00
           2       0.91      0.08
```

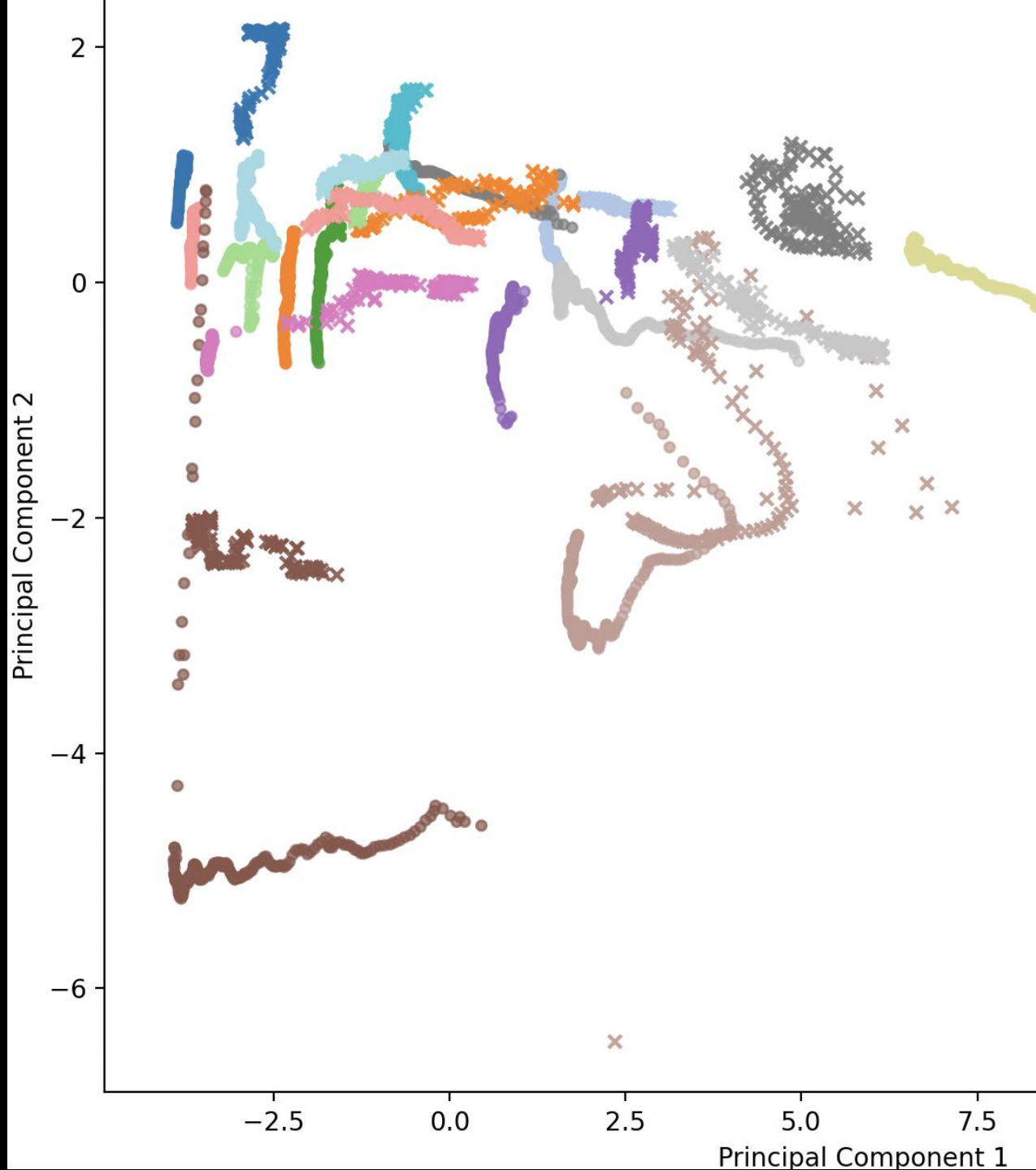
SEPARABILITY

- Significant Overlap
- First 2 components capture 99% of variance
- Captures true data structure (minimal information loss)
- Stress-related patterns are <u>not</u> cleanly separable in feature space
- Explains consistent ~65% classification performance



INDIVIDUAL VARIABILITY

- Some subjects exhibit clearer stress separation
- Others show significant overlap
- Indicates variability in physiological stress responses
- Global models may fail because they assume consistent patterns across subjects
- Stress separability is subject-dependent
- The data structure is driven more by individual identity than by stress states



FEATURE REPRESENTATION COMPARISON

We tested different ways of summarizing signals (ECG, EDA, temperature) and found that our way of doing so misses important patterns over time

- **ECG** alone failed to detect stress (0% recall)
- **EDA** showed limited improvement in stress detection (~8% recall)
- **Combined** features did not improve performance and often collapsed to baseline-only predictions

Indicates that adding modalities **does not improve** separability, and that the current statistical features (mean, std) **do not capture** meaningful stress-related patterns

RISK-ORIENTED BEHAVIOR

- Models systematically **miss** high-risk stress states and overwhelmingly predict the baseline class of **not stressed**.
- Many models completely **fail** to detect stress (0% recall).
- Only about **8%** of stress cases were **correctly detected**, meaning that models would miss stress when it actually happens.
- **High accuracy** (~65%) is **misleading**, and very likely to be driven by baseline dominance in the dataset.

Therefore, it is safe to deduce that standard classifiers are **not suitable** for risk-sensitive stress detection

WE FOUND THAT

- Physiological stress and baseline patterns show **strong overlap**, meaning that stress and non-stress signals look very similar in this dataset.
- Our models could not reliably tell these signals apart.
- Even with multiple features, stress remains **difficult to detect.**
- The limitation is not the model but the way the data is represented.
  - Simple statistical summaries (mean, std) are insufficient for accurately capturing stress indicators, and we may be **oversimplifying** complex biological responses.
  - The variation between individuals is stronger than the variation caused by stress

WE CAN CONCLUDE BASED ON THESE FINDINGS THAT STRESS DETECTION IS HIGHLY **SUBJECT-DEPENDENT** AND NOT GLOBALLY SEPARABLE

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Any Questions?